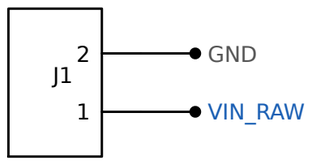
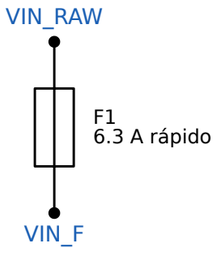
Cubo AP · Placa de control v3 (ESP32-C3 soldado) — esquema

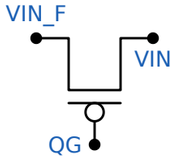
Generado de diseno.py (la misma lista de conexiones que la placa). Las etiquetas azules son nombres de red: pines con la misma etiqueta están conectados. Sin probar en hardware.

1 · Entrada del cartucho y protección

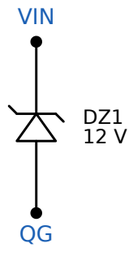


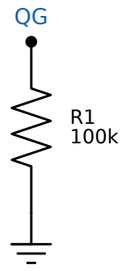
Entrada 12–36 V

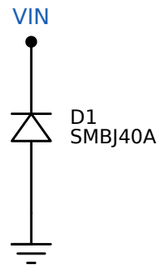


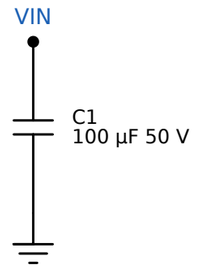


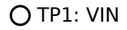
Q1 · P-MOSFET



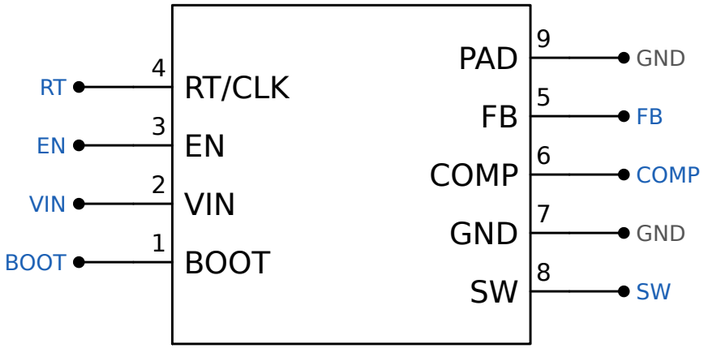




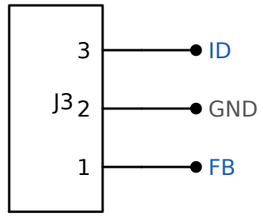




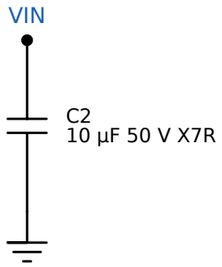
Q1 protege contra inversión de polaridad (el diodo interno conduce y luego el canal se enciende). DZ1 limita Vgs a 12 V. D1 recorta transitorios del cartucho.

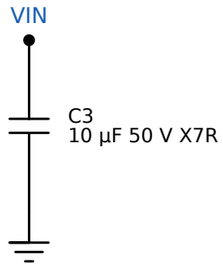


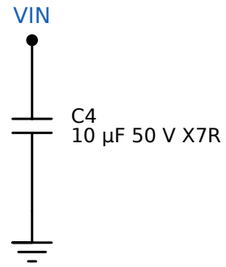
U1 · TPS54560DDA

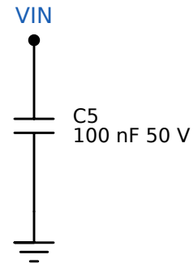


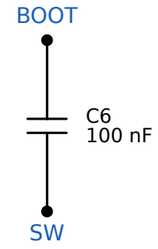
Tarjeta

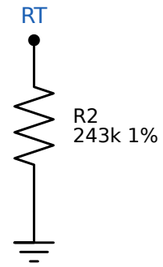


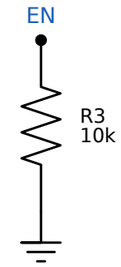


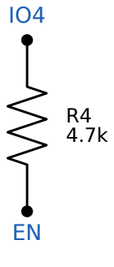


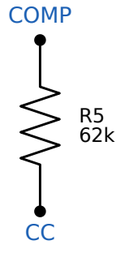


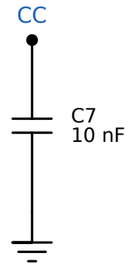


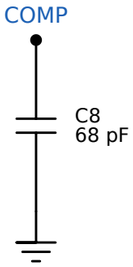


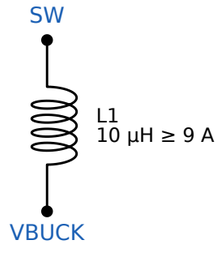


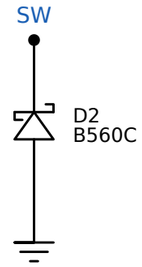


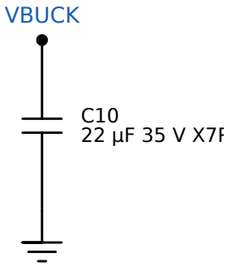


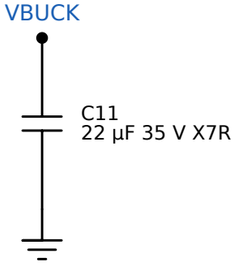


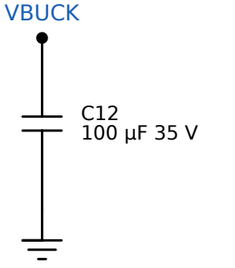


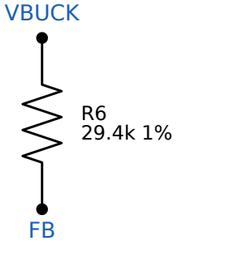


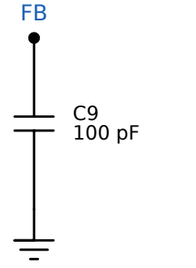


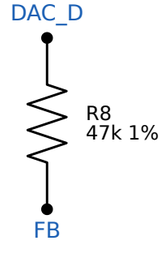


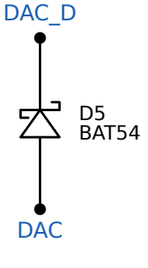


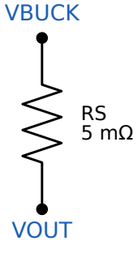


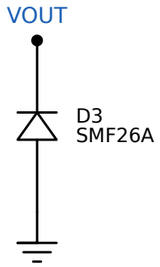






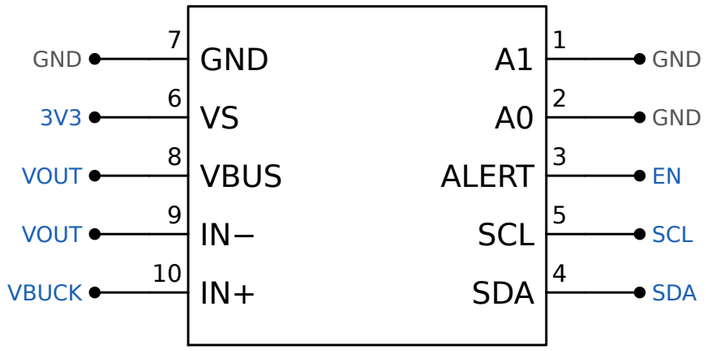




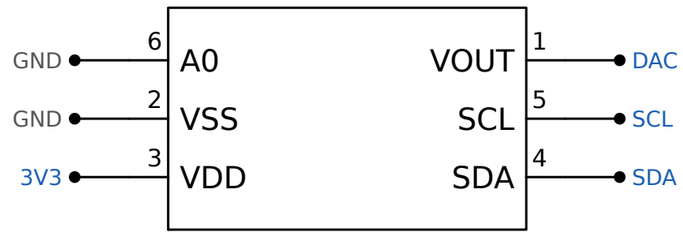


Vout = 0.8 V × (1 + 29.4k / R_B). R_B vive en la tarjeta (J3 pin 1–2): sin tarjeta, FB = Vout y la salida se cae a ~0.8 V. El DAC solo inyecta corriente al FB a través de D5: solo puede BAJAR el voltaje. Compensación tipo II para cruce ≈ 15 kHz a 12 V (verificar con la herramienta de TI). RS = 5 mΩ, sentido Kelvin al INA226.

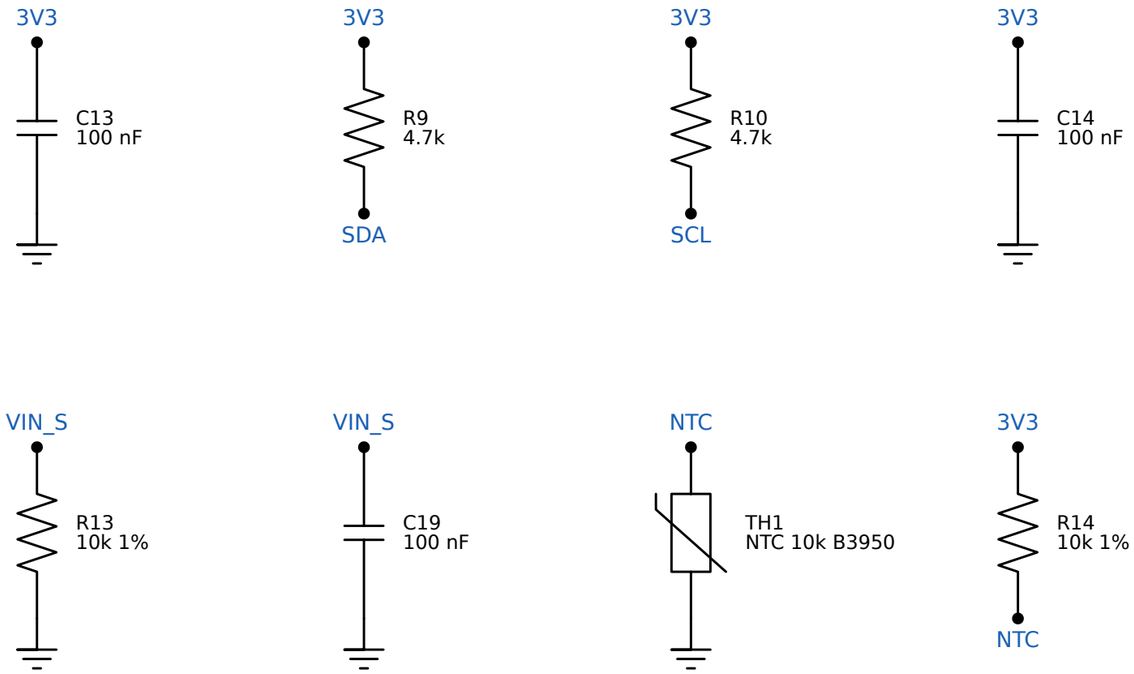
3 · Medición y ajuste: INA226, DAC MCP4725, ID, Vin y temperatura



U4 · INA226

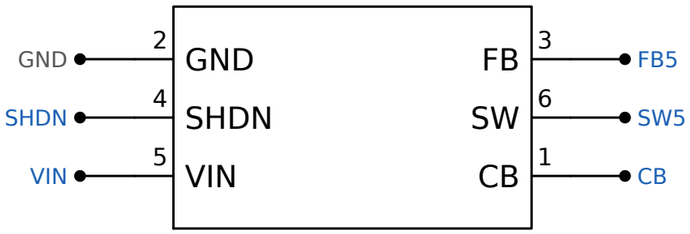


U5 · MCP4725A0

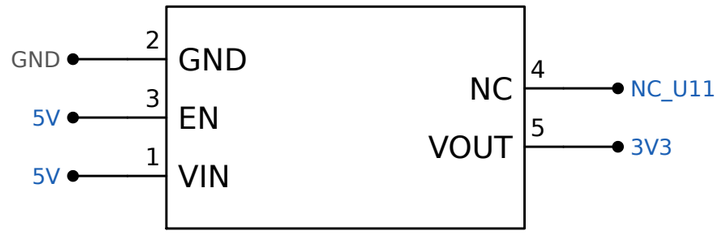


INA226 en 0x40 (A0 = A1 = GND), DAC en 0x60. ALERT del INA226 está en la red EN: al dispararse tira el EN del reductor a GND por hardware (~1 ms). Pull-up de ID = 47k (R_ID de la tarjeta: 10k…150k).

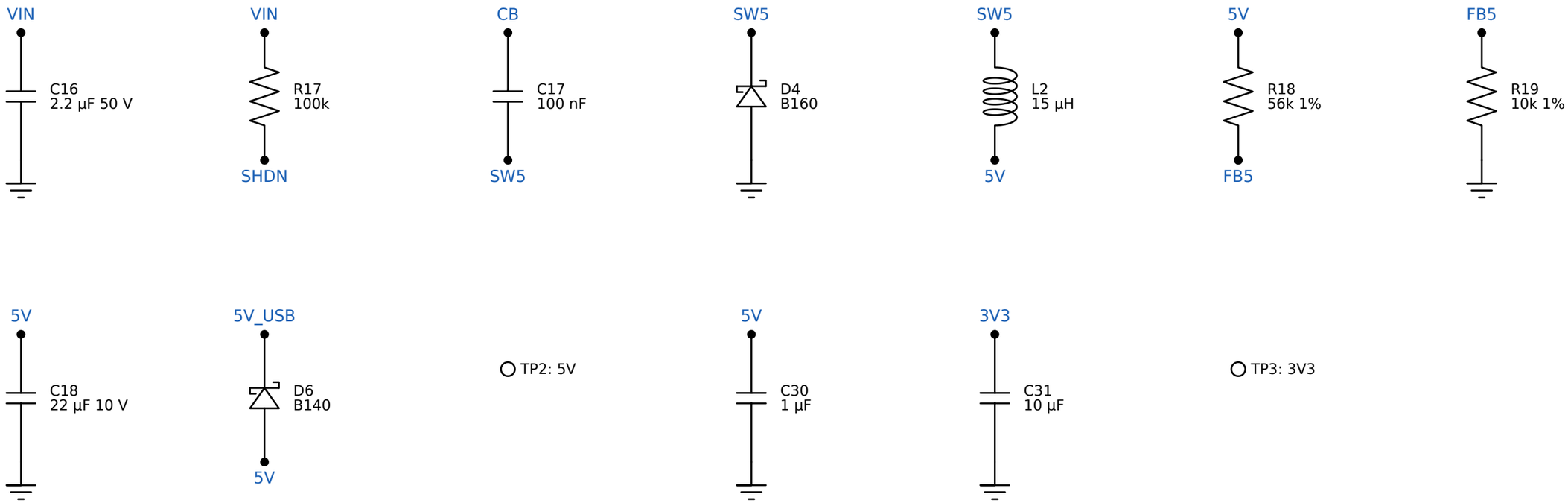
4 · 5 V auxiliar (LMR14006) y 3.3 V (AP2112K)



U2 · LMR14006

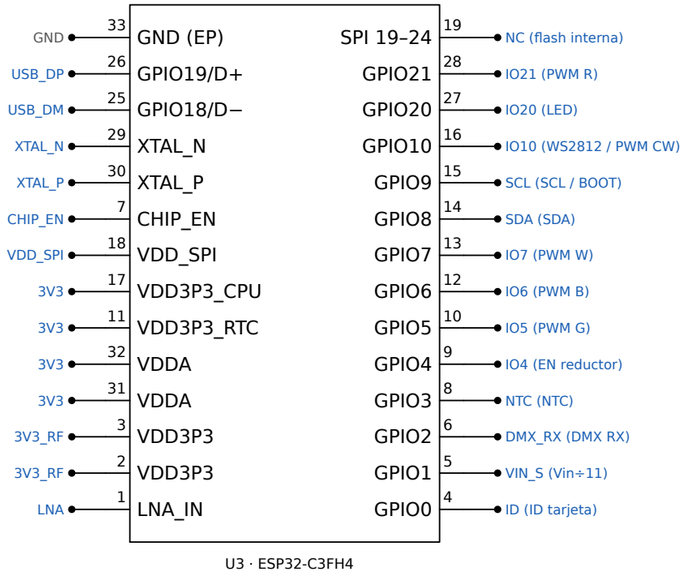


U11 · AP2112K-3.3



5 V para drivers y buffer WS2812; el AP2112K-3.3 (600 mA) da 3.3 V al ESP32-C3 y a los sensores (antes lo hacía el módulo). D6 evita alimentar al USB de la PC al programar con la placa conectada.

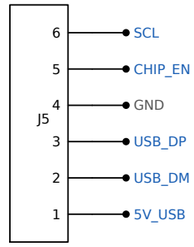
5 · ESP32-C3FH4 soldado: alimentación, reset, cristal, RF, programación y LED



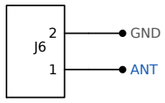
U3 · ESP32-C3FH4



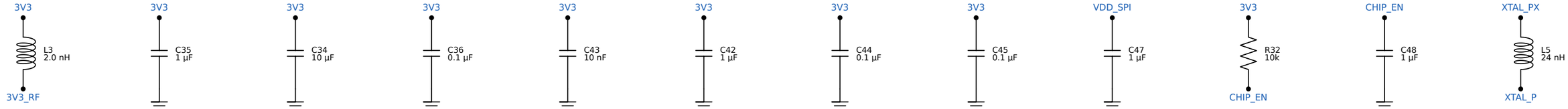
Y1 · 40 MHz

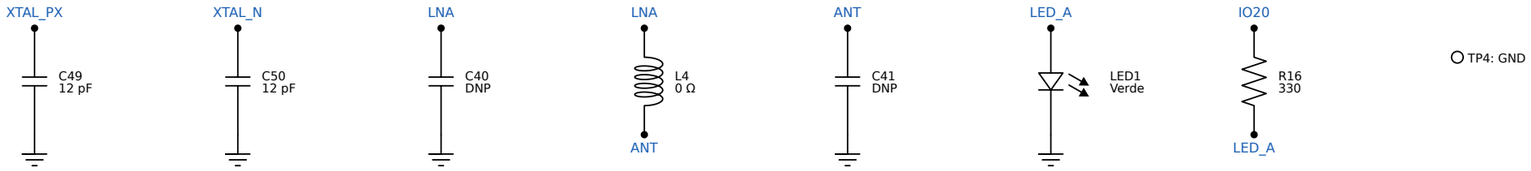


PROG



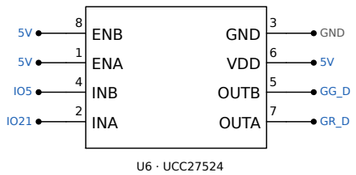
U.FL



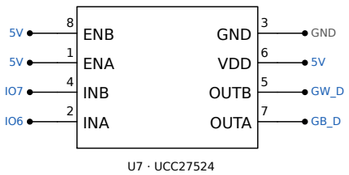


Chip ESP32-C3FH4 soldado (sin módulo), circuito según la guía de hardware de Espressif: VDD3P3 con 2.0 nH + 1/10/0.1 µF, VDDA 1 µF + 10 nF, RTC/CPU 0.1 µF, VDD_SPI 1 µF (alimenta la flash interna), CHIP_EN con RC 10k/1 µF. Cristal 40 MHz ±10 ppm con 24 nH en serie en XTAL_P y cargas de 12 pF (C = 2·(CL − Cparásita), ajustar con prueba de tono hasta ±10 ppm). RF: red CLC (C40/L4/C41) junto al chip, por sintonizar con analizador de redes; arranca con L4 = 0 Ω y C40/C41 sin poblar; tramo de 2 mm hasta el U.FL (0.8 mm con GND a 0.2 mm: coplanar ≈ 57 Ω en FR4 de 1.6 mm, muy corto frente a λ) y antena externa. Programación por el USB nativo del C3 (pads pogo J5); IO9 a GND durante el reset = modo descarga. Mismo mapa de GPIO que la v2: el firmware de la v2 corre sin cambios.

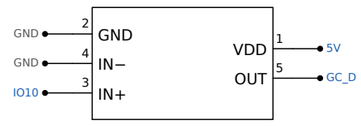
6 · Cinco canales PWM (UCC27524 ×2 + UCC27517 + MOSFET), datos WS2812 y salida



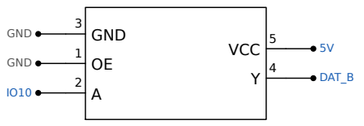
U6 · UCC27524



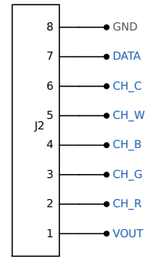
U7 · UCC27524



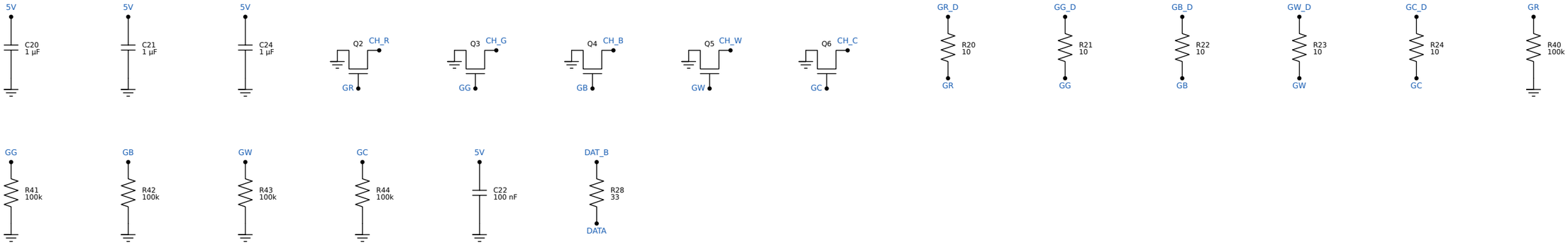
U10 · UCC27517



U8 · 74AHCT1G125

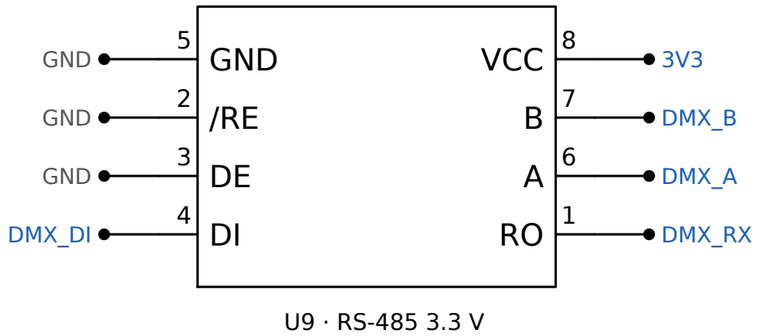


Salida LED

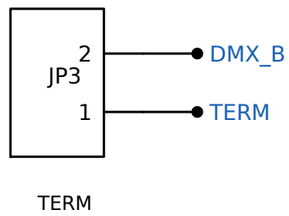


5.º canal (blanco frío) con UCC27517; su entrada comparte IO10 con los datos WS2812: el firmware no deja activar tiras RGB+CCT y direccionables a la vez. Drivers de 5 A: conmutación de ~20 ns a 19 kHz, sin calentar los MOSFET. R20–R24 amortiguan la compuerta; R40–R44 la mantienen apagada si falta 5 V. Ánodo común (V+) para tiras RGBW; GND y DAT para tiras direccionables.

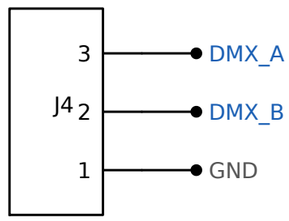
7 · DMX512 por cable (RS-485, solo recepción)



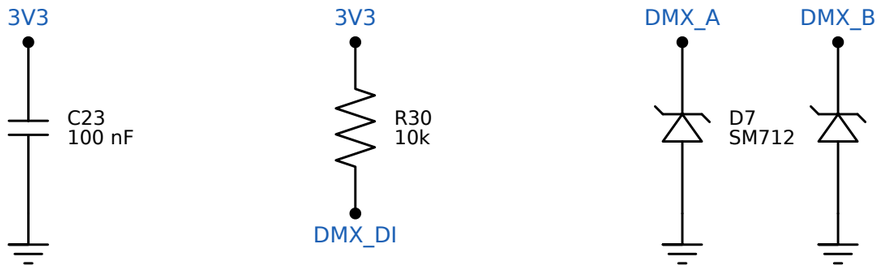
U9 · RS-485 3.3 V



TERM



DMX



Transceptor fijo en recepción (DE y /RE a GND, DI a 3.3 V). SM712 protege A/B contra descargas (−7/+12 V). La terminación de 120 Ω solo se pone (JP3) en el último equipo. Bornera: 1 GND, 2 D−, 3 D+ = pines 1, 2 y 3 del XLR.